

Gravity

The mass of gas in a star is held in place by gravity. A star is said to be self-gravitating when it contains sufficient mass for its own gravitational force-field to be sufficiently strong to keep itself together.

The material of a star is acted on by two forces: the gravitational force, which tends to draw the stellar material inward, and a pressure force which pushes outward. In most stars, these two forces are exactly balanced at every point of the star. An imbalance leads to a net acceleration or motion, or in the case of a large imbalance-- a super-novae explosion. If the pressure support is removed, the star would collapse under its own gravity. Most stars with equal pressure and gravitational forces are said to be in 'hydrostatic equilibrium'.

Gravitation can convert an innocuous rotating variable into an exotic one, by amplifying both the rotation and the magnetic field. Gravitation dominates the last phases of a star's evolution, converting it into a compact white dwarf, an even more compact neutron star, or an infinitely dense, compact black hole.

The behavior of the universe as a whole is dominated by gravity, as are the motions of the planets. It is one of the four fundamental fields. Gravity is the dominant force over astronomical distances.

Newton's law of gravity:

$$\text{constant 'G' } \times \frac{(\text{mass of one body}) \times (\text{mass of another body})}{(\text{distance between bodies})^2}$$

The two bodies thus exert an equal and opposite force on each other. 'G' is called the gravitational constant.

The escape velocity is the minimum velocity which one body must have to escape from another's gravitational attraction.

General relativity is the currently accepted gravitational theory, as stated below:

$$\text{Energy} = \text{Mass} \times (\text{speed of light})^2$$

And, to get down to earth, I find it equally interesting that all in nature is balanced, be it plants, animals or ourselves.

by Harold Moorin

May (taken from Astronomical Calendar 1989, by Guy Ottewill)

- 1 Mon May Day. Phi Bootid and Alpha Scorpid Meteors.
(3 UT) Mercury at greatest elongation east, 20.8 degrees directly above sun as sunset, "The best view all year!"
- 4 Thr (5 UT) Moon at perigee. Distance 56.4 earth-radii.
(7 UT) Pluto at opposition.
- 5 Fri (11:47 UT) New moon.
- 6 Sat (22 UT) Moon passes 3 degrees north of Mercury.
- 7 Sun (7 UT) Moon passes 5 degrees north of Jupiter.
- 9 Tue (1 UT) Moon passes 3 degrees north of Mars.
- 10 Wed (19 UT) Mercury and Jupiter nearest to each other (5.5).
- 12 Fri (14:49 UT) Moon at 1st quarter.
(23 UT) Mercury begins retrograde motion (westward).
- 13 Sat (7 UT) Venus at ascending node through ecliptic plane.
(9 UT) Moon at descending node (longitude 151.3).
(21 UT) Sun enters Taurus, longitude 151.3 on ecliptic.
- 16 Tue (7 UT) Mercury 0.6 degrees north of Venus; 11 degrees from the sun in the evening sky; magnitudes 3.4 & -3.9
(9 UT) Moon at apogee. Distance 63.6 earth-radii.
- 18 Thr Omicron Cetid meteors (daytime).
(0 UT) Asteroid Juno passes 15' south of Regulus.
(19 UT) Mercury at descending node through ecliptic.
- 19 Fri (19 UT) Venus 5.8 degrees north of Aldebaran. Both 12 degrees from sun in the evening sky.
- 20 Sat Full moon, called Planting Moon or Milk Moon.
- 23 Tue (4 UT) Venus 0.8 degrees north of Jupiter; 13 degrees from sun in the evening sky.
(12 UT) Moon passes 4 degrees south of Uranus.
(22 UT) Mercury at inferior conjunction with sun.
- 24 Wed (1 UT) Moon passes 5 degrees south of Neptune.
(4 UT) Moon passes 4 degrees south of Saturn.
- 27 Sat (15 UT) Moon at ascending node (longitude 329.6).
- 28 Sun (4 UT) Moon at last quarter.
- 29 Mon (3 UT) Mercury at aphelion, 0.4667 a.u. from sun.
- 30 Tue (6 UT) Moon passes 0.7 degrees south of Pallas.

Editor's notes

At this month's meeting will be several reservation forms for the attendance of this year's Riverside Amateur Telescope Makers Conference. If reservations are made before May 1st, the fee is less than if after the above date.

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The ballot below presents the current nominations presented by the Phoenix Astronomical Society nominating committee for the 1989-1990 year. If there are any other nominations than the ones presented below, please fill them in the space provided.

 (please bring completed, Friday the 28th of May)

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